PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.
AUTHORS	Tesfay, Fisaha; Javanparast, Sara; Gesesew, Hailay; Mwanri, Lillian; Ziersch, Anna

VERSION 1 – REVIEW

REVIEWER	Todowede, Olamide University of York, Department of Health Science
REVIEW RETURNED	22-Dec-2020

GENERAL COMMENTS	This is an interesting study, and the authors have stated a clear objectives. This study is required within the population, understanding the increasing issues of food security and malnutrition. However, the study requires editing and more structure. I will believe that further analysis can be done for studies that reported BMI to indicate if nutritional programs have an impact on the BMI of people infected with HIV in SSA. In this instance the author has not addressed this issue, The study results requires more concise reporting of the results, to replace the present form - presentations of tables this seems overwhelming.

			Reviewers Comments		
		1	Data sources: We searched for primary studies published in various databases, google, google scholar and reference of refences.	Author should look at the databases searched again and align the reporting. I doubt the possibilities of searching of google as stated in the Abstract by the author. What about Pubmed, Ebscohost. Author should synchronise that Data sources stated in the main manuscript with what is in the Abstract.	
		2	……..programs are different from other settings that are why the current review is limited to studies conducted in sub-Saharan Africa only.	Authors should please reword this sentence, I will think …..that is why this review is ……, will be more appropriate	
		3	This is a systematic review and it is nor applicable to make patient and public involvement.	The manuscript requires editing	
		4	A total of 2006 and 30 papers respectively were obtained from electronic databases and through other sources.	This sentence is not clear, do you mean 2036 articles were found.	

		5	Data Analysis	I would have thought that a meta-analysis be done for 9 studies that reported BMI, there are methods to treat heterogeneity - https://handbook-5-1.cochrane.org/chapter_9/9_5_3_strategies_for_addressing_heterogeneity.htm I do not think the author has made effort to treat the heterogeneity, before concluding to do a narrative synthesis. I would have thought the author will make use of the BMI's reported, since most study reported this measure.	
		6	In one study, not being on ART was associated with nutritional recovery [40]. the viral load significant negative correlation with the lean body mass and BMI (statistics not reported)	Please check this statements and report accordingly based on what was reported in the article	

REVIEWER	Derose, Kathryn RAND Corp
REVIEW RETURNED	29-Jan-2021

GENERAL COMMENTS	This manuscript conducts a systematic review of nutrition programs focusing on undernutrition among people living with HIV in Sub-Saharan Africa. The study is important because nutrition plays an important role in HIV disease progression and international bodies have called for the integration of nutritional programs into HIV care and support programs. The following are suggestions on how to improve the manuscript's contribution to the literature: 1. The manuscript states as its purpose "to synthesise the existing evidence on the impact of nutritional programs on nutritional outcomes of people living with enrolled in nutritional countries in

	HIV care" (p. 4). There are several issues regarding this statement. a. There is something wrong with this sentence (words missing? Extra words?) b. The review is focused on nutrition programs addressing undernutrition among people with HIV and this needs to be made clearer throughout the manuscript, since nutrition programs can also address overnutrition or overweight and obesity. c. The statement of purpose should also mention the geographic focus in sub-Saharan Africa. 2. The manuscript identifies the key defining characteristics of nutritional programs in HIV care as "nutrition assessment, counseling and support services" (p. 4). However, it does not appear that this review considered the first two, nutrition assessment and counseling, as only nutrition supports are described under "Characteristics of the nutritional programs" (pp. 7-8). The extent which nutritional assessment was done and how are important to report in Table 3 and the text, as well as any types of nutritional counseling done (and by whom). These are key features of nutritional programs – knowing whether they have been implemented along with food/nutrition supplementation and how is important for the field. 3. The scoring for Supplementary file 3 (Assessment of methodological quality) is confusing. a. The scoring should be better explained in the paper. b. All the non-Y answers for Q2 are listed as "N/" – is this supposed to be NA? c. Q3 has N/A and N/ - but key only gives "NA" d. Many that scored "Y" on all questions are listed as "10" – should these be 100%? 4. This review focuses only on nutritional programs that address undernutrition, however, another form of malnutrition affecting people with HIV, including in sub-Saharan Africa, is overweight and obesity. For example, see Coetzee et al. HIV, antiretroviral therapy and non-communicable diseases in sub-Saharan Africa: empirical evidence from 44 countries over the period 2000 to 2016. I Int AIDS Soc 2019. ART and an ageing HIV-positive population bring increased cardiovascular risks (e.g., diabetes, overweight). Although this is not the focus of the current review, the trend should at least be mentioned in the Discussion as a nutritional issue facing the field for which programs will need to be developed. 5. Other limitations should be considered, e.g., publications in languages other than English.

VERSION 1 – AUTHOR RESPONSE

	Reviewer 1 (Dr. Olamide Todowede, University of York)		
	General		
	This is an interesting study, and the authors have stated a clear objective. This study is required within the population, understanding the increasing issues of food security and malnutrition. However, the study requires editing and more structure. I will believe that further analysis can be done for studies that reported BMI to indicate if nutritional programs have an impact on the BMI of people infected with HIV in SSA. In this instance the author has not addressed this issue, the study results requires more concise reporting of the results, to replace the present form - presentations of tables this seems overwhelming.	Thank you for the feedback. The overall objective of this review is to synthesis the existing evidence on the impact of nutritional programs on the nutritional status of people living with HIV. BMI is used in some instances to measure the impact of the nutritional programs on the nutritional status of people living with HIV. However, there were significant variations in the definition of the various nutritional indicators in various program such nutritional recovery, failure, non-response, default. Hence, we believe that conducting meta-analysis on these parameters will not be meaningful. We have moved the tables to an annex to simplify the main paper.	N/A

1	Data sources: We searched for primary studies published in various databases, google: Author should look at the databases searched again and align the reporting. I doubt the possibilities of searching of google as stated in the Abstract by the author. What about Pubmed, Ebscohost. Author should synchronise that Data sources stated in the main manuscript with what is in the Abstract.	We have now included in the abstract all databases that we searched. This included Medline which is a subset of Pubmed.	Page 2 and 4
2	……..programs are different from Other settings that are why the current review is limited to studies conducted in sub- Saharan Africa only: Authors should please reword this sentence, I will think …..that is why this review is ……, will be more appropriate	We appreciate your suggestion and have addressed it.	Page 6
3	This is a systematic review and it is not applicable to make patient and public involvement: The manuscript requires editing	Thank you for nothing that: Revised accordingly.	
4	A total of 2006 and 30 papers respectively were obtained from electronic databases and through other sources: This sentence is not clear; do you mean 2036 articles were found.	Yes, we have revised it accordingly.	Page 7
5	Data Analysis: I would have thought that a meta-analysis be done for 9 studies that reported BMI, there are methods to treat heterogeneity - https://handbook-5-1.cochrane.org/chapter_9/9_5_3_strategies_for_addressing_heterogeneity.htm I do not think the author has made effort to treat the heterogeneity, before concluding to do a narrative synthesis. I would have thought the author will make use of the BMI's reported, since most study reported this measure.	Thank you. We are aware of the method to handle heterogeneity but as we explained it above the parameters used by various programs and the type of nutritional support given in various countries is also different. The other issues is that the raw BMI is not reported in each study rather it was given as an aggregate measure (e.g. recovery is defined BMI>18.5, 18 and 20kg/m^2) but at what BMI does each patient recovered was not given except	Page 7

		the aggregate reports of recovery, failure and default from the nutritional programs	
	In one study, not being on ART was associated with nutritional recovery 40. the viral load significant negative correlation with the lean body mass and BMI (statistics not reported: Please check this statements and report accordingly based on what was reported in the article	Revised accordingly. That is how it is reported in the primary study.	Annex 1
	Reviewer 2 (Dr. Kathryn Derose, RAND Corp)		
	This manuscript conducts a systematic review of nutrition programs focusing on undernutrition among people living with HIV in Sub-Saharan Africa. The study is important because nutrition plays an important role in HIV disease progression and international bodies have called for the integration of nutritional programs into HIV care and support programs. The following are suggestions on how to improve the manuscript's contribution to the literature:	Thank you for your comments.	
1	1. The manuscript states as its purpose "to synthesise the existing evidence on the impact of nutritional programs on nutritional outcomes of people living with enrolled in nutritional countries in HIV care" (p. 4). There are several issues regarding this statement. a) There is something wrong with this sentence (words missing? Extra words?) b) The review is focused on nutrition programs addressing undernutrition among people with HIV and this needs to be made clearer throughout the manuscript, since nutrition programs can also address overnutrition or overweight and obesity c) The statement of purpose should also mention the geographic focus in sub-Saharan Africa.	a) Thank you. Yes, we have revised this sentence. b) Revision has made to clarify the ultimate focus of this paper on undernutrition and the relevance of programs to address overnutrition and other factors included as an area for future research in the strengths and limitations section. c) The geographic focus of this review is in sub-Saharan	Page 4

		Africa and we have highlighted that in the statement of purpose.	
2	The manuscript identifies the key defining characteristics of nutritional programs in HIV care as “nutrition assessment, counseling and support services” (p. 4). However, it does not appear that this review considered the first two, nutrition assessment and counseling, as only nutrition supports are described under “Characteristics of the nutritional programs” (pp. 7-8). The extent which nutritional assessment was done and how are important to report in Table 3 and the text, as well as any types of nutritional counseling done (and by whom). These are key features of nutritional programs – knowing whether they have been implemented along with food/nutrition supplementation and how is important for the field.	We have added the missing components of a nutritional program which includes nutritional assessment and counselling in the tables and in the text. Nutritional assessment was done in all studies, but nutritional counseling was stated only in two studies. We have included some information on nutritional assessment and counseling on the characteristics of the nutritional. However, including the extent of nutritional assessment and counseling and who did it would be out of the aim of this review. The aim of this review is to determine the characteristics and impacts of nutritional programs on the nutritional status of people living with HIV.	Page 8

3	The scoring for Supplementary file 3 (Assessment of methodological quality) is confusing. a) The scoring should be better explained in the paper. b) All the non-Y answers for Q2 are listed as "N/" – is this supposed to be NA? c) Q3 has N/A and N/ - but key only gives "NA" d) Many that scored "Y" on all questions are listed as "10" – should these be 100%?	a) We have revised the explanation of the scoring. b) Apologies, this was due to the misalignment of the table. We have adjusted the table now. c) We were referring to N/A and this happened due the misalignment of the table. We have adjusted it now. d) Same as above. We meant 100 and this happened due the misalignment of the table. We have addressed it now.	N/A
4	This review focuses only on nutritional programs that address undernutrition, however, another form of malnutrition affecting people with HIV, including in sub-Saharan Africa, is overweight and obesity. For example, see. • Coetzee et al. HIV, antiretroviral therapy and non-communicable diseases in sub-Saharan Africa: empirical evidence from 44 countries over the period 2000 to 2016. I Int AIDS Soc 2019.	Thank you for highlighting this. As above, we have stressed that this review is focused on nutritional programs that address	Page 16

	• ART and an ageing HIV-positive population bring increased cardiovascular risks (e.g., diabetes, overweight). Although this is not the focus of the current review, the trend should at least be mentioned in the Discussion as a nutritional issue facing the field for which programs will need to be developed.	undernutrition. We have highlighted the importance of also considering overnutrition in the discussion.	
5	Other limitations should be considered, e.g., publications in languages other than English.	Thank you. We have added further limitations including in relation to English.	

VERSION 2 – REVIEW

REVIEWER	Derose, Kathryn RAND Corp
REVIEW RETURNED	20-Apr-2021

GENERAL COMMENTS	The authors have addressed most of my previous concerns, however, the following issues remain: 1. The authors have done some work to clarify that this review focuses only on undernutrition among people with HIV, however, there are still numerous places where the authors refer to "malnutrition" in the text and tables as if this only meant undernutrition. For clarity, it would be good to change the wording to "undernutrition" given that malnutrition can also mean overweight, per the WHO definition of malnutrition: Malnutrition refers to deficiencies, excesses, or imbalances in a person's intake of energy and/or nutrients. The term malnutrition covers 2 broad groups of conditions. One is 'undernutrition'—which includes stunting (low height for age), wasting (low weight for height), underweight (low weight for age) and micronutrient deficiencies or insufficiencies (a lack of important vitamins and minerals). The other is overweight, obesity and diet-related noncommunicable diseases (such as heart disease, stroke, diabetes, and cancer). 2. Given the above definition, micronutrient deficiencies would be another type of undernutrition. Were these outcomes not examined in the literature, or were the authors only interested in weight-related undernutrition? This should be clarified in the manuscript. 3. The authors refer several times to "nutritional outcomes," but it appears these only include weight-related under nutrition (underweight). For clarity, the authors should define what they included under "nutritional outcomes" and if it was always underweight, consider using this as the term throughout the paper (or "weight-related nutritional outcomes").

	4. The authors added a statement to the end of Strengths and Limitations about how overnutrition has been highlighted as an important issue for people living with HIV and cite Aberman et al. 2014. This does not seem like the best citation for this statement, as Aberman et al. 2014 focus on food security and nutrition interventions in response to the AIDS epidemic (nutrition supplementation for undernourished people with HIV, safety nets, and livelihood interventions) – they don't talk much if at all about overnutrition or overweight. There are many citations about the increasing risk for overweight and cardiovascular disease among people with HIV on long-term ART treatment. Here are two review articles from sub-Saharan Africa: Coetzee et al. HIV, antiretroviral therapy and non-communicable diseases in sub-Saharan Africa: empirical evidence from 44 countries over the period 2000 to 2016. I Int AIDS Soc 2019. Achwoka D, Waruru A, Chen TH, Masamaro K, Ngugi E, Kimani M, Mukui I, Oyugi JO, Mutave R, Achia T, Katana A, Ng'ang'a L, De Cock KM. Noncommunicable disease burden among HIV patients in care: a national retrospective longitudinal analysis of HIV-treatment outcomes in Kenya, 2003-2013. BMC Public Health. 2019 Apr 3;19(1):372. 5. The Abstract should be edited to reflect the focus on undernutrition – e.g., a. This systematic review sought to synthesise evidence on the characteristics and impact of nutritional programs on underweight among people living with HIV in sub-Saharan Africa. b. Nutritional programs in HIV care have led to some improvements in undernutrition amongst people living with HIV. However, the programs were characterised by a high magnitude of default and non-response. To improve underweight among people living with HIV, a holistic approach that addresses longer-term determinants of undernutrition is needed.

VERSION 2 – AUTHOR RESPONSE

	Reviewer 2 (Dr. Kathryn Derose, RAND Corp)		
1	The authors have done some work to clarify that this review focuses only on undernutrition among people with HIV, however, there are still numerous places where the authors refer to "malnutrition" in the text and tables as if this only meant undernutrition. For clarity, it would be good to change the wording to "undernutrition" given that malnutrition can also mean overweight, per the WHO definition of malnutrition: Malnutrition refers to deficiencies, excesses, or imbalances in a person's intake of energy and/or nutrients. The term malnutrition covers 2 broad groups	Thank you for the feedback and we have addressed it accordingly. We have changed the term "malnutrition" to "undernutrition" throughout.	

	of conditions. One is 'undernutrition'—which includes stunting (low height for age), wasting (low weight for height), underweight (low weight for age) and micronutrient deficiencies or insufficiencies (a lack of important vitamins and minerals). The other is overweight, obesity and diet-related noncommunicable diseases (such as heart disease, stroke, diabetes, and cancer).		
2	Given the above definition, micronutrient deficiencies would be another type of undernutrition. Were these outcomes not examined in the literature, or were the authors only interested in weight-related undernutrition? This should be clarified in the manuscript.	Thank you very much. We only focused on weight related undernutrition. Accordingly, we have clarified this in the first paragraph of [Introduction] "Undernutrition involves weight related (stunting (low height for age), wasting (low weight for height), underweight (low weight for age) and micronutrient deficiencies but this particular review will focus on weigh related undernutrition".	Page 3
3	The authors refer several times to "nutritional outcomes," but it appears these only include weight-related under nutrition (underweight). For clarity, the authors should define what they included under "nutritional outcomes" and if it was always underweight, consider using this as the term throughout the paper (or "weight-related nutritional outcomes").	In the present study, weight-related nutritional outcomes refer to recovery, program failure, non-response and default. For clarity purposes, we have described as positive nutritional outcomes when we refer to "recovery", and negative nutritional outcomes when we refer to otherwise. We have added this in the revised document.	
4	The authors added a statement to the end of Strengths and Limitations about how overnutrition has been highlighted as an important issue for people living with HIV and cite Aberman et al. 2014. This does not seem like the best citation for this statement, as Aberman et al. 2014 focus on food security and nutrition interventions in response to the AIDS epidemic (nutrition supplementation for undernourished people with HIV, safety nets, and livelihood interventions) – they don't talk much if at all about overnutrition or overweight. There are many citations about the increasing risk for overweight and cardiovascular disease among people with HIV on long-term ART treatment. Here are two review articles from sub-Saharan Africa:	Thank you very for the recommendation of these relevant citations. We have replaced the earlier citation with other relevant citations, including your suggestions.	Page 16

	1. Coetzee et al. HIV, antiretroviral therapy and non-communicable diseases in sub-Saharan Africa: empirical evidence from 44 countries over the period 2000 to 2016. I Int AIDS Soc 2019. 2. Achwoka D, Waruru A, Chen TH, Masamaro K, Ngugi E, Kimani M, Mukui I, Oyugi JO, Mutave R, Achia T, Katana A, Ng'ang'a L, De Cock KM. Noncommunicable disease burden among HIV patients in care: a national retrospective longitudinal analysis of HIV-treatment outcomes in Kenya, 2003-2013. BMC Public Health. 2019 Apr 3;19(1):372.		
5	The Abstract should be edited to reflect the focus on undernutrition – e.g., a. This systematic review sought to synthesise evidence on the characteristics and impact of nutritional programs on underweight among people living with HIV in sub-Saharan Africa. b. Nutritional programs in HIV care have led to some improvements in undernutrition amongst people living with HIV. However, the programs were characterized by a high magnitude of default and non-response. To improve underweight among people living with HIV, a holistic approach that addresses longer-term determinants of undernutrition is needed.	We have revised the abstract to reflect weight related nutritional outcomes. Please see above in comment #3 for the description of nutritional outcomes.	